

UNIVERSIDAD DE LAS FUERZAS ARMADAS ESPE

DEPARTAMENTO DE CIENCIAS DE LA COMPUTACIÓN

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DESARROLLO WEB AVANZADO

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APELLIDOS Y NOMBRES: Angel Sabando, David Rodriguez, Esteban Quiroga

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This technical report documents the agreements and strategic vision established for the development of the "Accounting System - ATS Generator." The project, led by Esteban Quiroga, Angel Sabando, and David Rodriguez, aims to transcend simple office automation to become a robust software engineering solution capable of handling current tax complexities.

Introduction and Project Context

The meeting focused on the urgent need to optimize the interaction between taxpayers and the Internal Revenue Service (SRI). As discussed during the technical session, the manual process of downloading invoices one by one—citing a case with 41 receipts in a single month—is inefficient and prone to human error. The team analyzed the "Gestor Plus" system, a Java-based tool currently used to capture XML files from the SRI portal, which serves as a benchmark for the functionalities the new system must integrate and improve.

Analysis of Features and Technical Challenges

The core of the project is divided into critical modules that ensure the integrity of financial data:

- **Data Integration and Management:** The system will implement a secure connection with the "SRI online" portal. This functionality will allow for the bulk download of invoices filtered by tax periods (monthly or semi-annually), eliminating the manual workload demonstrated during the review of the XML and RIDE (PDF) capture process.
- **Generation of the Simplified Transactional Annex (ATS):** One of the pillars is the automatic creation of the ATS matrix in XLSM format. As observed in the data flow demonstration, the system must parse the invoice to extract fields such as RUC, tax base, VAT, and authorization numbers, organizing them into the technical structure required by the SRI.
- **Validation and Export:** The system does not merely collect data; it acts as a quality filter. Before the final export to XML format (compatible with the DIMM SRI), a field validation will be executed to ensure the file complies with UTF-8 encoding and current business rules.

Conclusion

The "Accounting System" project is poised to be an essential tool for the modern accountant. By integrating process traceability (FT-06), the system will allow auditors and administrators to visualize every stage of the workflow, from initial linking to the successful generation of the annex. This initiative, framed within the Advanced Web Development course, represents a significant step forward in the digitalization of fiscal processes in Ecuador, ensuring security through HTTPS/TLS 1.2+ protocols in all communications.